Sevgi Gokce Kafali, Ph.D.

Alexander von Humboldt Post-doctoral Fellow

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RESEARCH INTERESTS

Focus Areas Medical Image Processing, Artificial Intelligence, Quantitative MRI, MR Image Analysis, MR Image Reconstruction

EXPERIENCE

10.2025 – present	**Alexander von Humboldt Post-doctoral Fellow** Technical University of Munich *(Host: Daniel Rueckert, Ph.D.)*	Munich, Germany
11.2024 – 09.2025	**Post-doctoral Researcher** Technical University of Munich *(Supervisor: Daniel Rueckert, Ph.D.)*	Munich, Germany
09.2018 – 09.2024	**Graduate Research Assistant** University of California, Los Angeles	Los Angeles, CA, USA
06.2021 – 09.2021	**Biomedical Engineering Ph.D. Intern** Siemens Healthineers	Los Angeles, CA, USA
02.2015 – 04.2018	**Graduate Research Assistant** Bilkent University	Ankara, Turkey

EDUCATION

07.2020 – 09.2024	**Ph.D. in Bioengineering** *Supervisor: Holden H. Wu, Ph.D.*	University of California, Los Angeles
09.2018 – 06.2020	**M.S. in Bioengineering** *Supervisor: Holden H. Wu, Ph.D.*	University of California, Los Angeles
09.2015 – 04.2018	**M.S. in Electrical and Electronics Engineering** *Supervisor: Emine Ulku Saritas, Ph.D.*	Bilkent University
09.2011 – 08.2015	**B.S. in Electrical and Electronics Engineering**	Bilkent University
03.2014 – 08.2014	**Exchange Student (Electrical & Electronics Eng.)**	Univ. of Erlangen-Nuremberg

THIRD-PARTY FUNDING

07.2026 – 07.2027	**Exchange Grant (UCLA-TUM collaboration)**, Bavarian California Technology Center (BaCaTech)
10.2025 – 10.2026	**Alexander von Humboldt Fellowship**, specifically awarded by the Global Minds Initiative, BMFTR Germany

ACHIEVEMENTS & AWARDS

2022	**Trainee Stipend (Monetary Award)**, ISMRM MRE Workshop
2022	**Pediatric Study Group Best Abstract Finalist**, ISMRM
2017, 2020, 2022	**Magna Cum Laude Award (top 15% abstracts)**, ISMRM (x4 across years)
2020	**Travel Grant Award**, IEEE ISBI
2018	**Bioengineering Department Fellowship**, University of California, Los Angeles
2016 – 2018	**Educational Stipend**, ISMRM
2015 – 2018	**100% Merit Scholarship & Monthly Stipend for MS studies**, Bilkent University
2011 – 2015	**50% Merit Scholarship for BS studies**, Bilkent University

SUPERVISED STUDENTS

09.2026 – pres.	**Boran Kilic**, M.S. Thesis *(co-supervised with Yundi Zhang)*
01.2026 – 07.2026	**Nithin Balan**, M.S. Thesis *(co-supervised with Niklas Bübeck)*
10.2025 – 03.2026	**Ruben Klinger**, M.S. Guided Research Project
08.2025 – pres.	**Sophie Starck**, Ph.D. *(co-supervised with Prof. Daniel Rückert)*
07.2025 – pres.	**Yundi Zhang**, Ph.D. *(co-supervised with Prof. Daniel Rückert)*
04.2025 – pres.	**Busra Nur Zeybek**, M.S. Thesis and Voluntary Research

TEACHING EXPERIENCE (SELECTED)

2026 (WS)	**AI in Medicine I and II**, Technical University of Munich *(Co-organizer)*
2025 (SS), 2026 (SS)	**Deep Learning for Inverse Problems in Medical Imaging** *(Organizer of a master's seminar)*, Technical University of Munich
2020 (SS), 2021 (SS)	**Advanced Topics in Magnetic Resonance Imaging** *(Teaching assistant of a graduate course)*, University of California, Los Angeles

PROFESSIONAL SERVICE & MEMBERSHIPS

Societies	International Society of Magnetic Resonance in Medicine (ISMRM), Medical Image Computing and Computer Assisted Interventions (miccai), IEEE Engineering in Medicine and Biology Society (EMBS)
Reviewer	Journal of Magnetic Resonance Imaging (JMRI), Scientific Reports (Nature), Journal of Quantitative Imaging in Medicine and Surgery (QIMS), NMR in Biomedicine, MICCAI (Main Conference & RIME Workshop), ISMRM Annual Meetings

PEER-REVIEWED JOURNAL PUBLICATIONS

2026 **SG Kafali**. Editorial for "High-Throughput Quantitative Chemical Shift-Encoded MRI of the Liver". *J Magn Reson Imaging*. DOI: 10.1002/jmri.70450

2026 **SG Kafali**. Editorial for "Measurements of Liver and Bone Marrow Proton Density Fat Fraction in the Setting of Ferumoxytol-Enhanced MRI". *J Magn Reson Imaging*. DOI: 10.1002/jmri.70534

2026 **SG Kafali***, W Zhang*, TR Adamos, K Kuwahara, A Dong, J Li, S Shih, TI Delgado, S Chowdhury, S Loong, J Moretz, SR Barnes, Z Li, S Ghahremani, KL Calkins, HH Wu. Cross-Cohort Federated Learning for Pediatric Abdominal Adipose Tissue Segmentation and Quantification Using Free-Breathing 3D MRI. *Radiology Advances*. DOI: 10.1093/radadv/umag002 (*co-first authors)

2024 **SG Kafali**, BD Bolster Jr., S Shih, TI Delgado, V Deshpande, X Zhong, TR Adamos, S Ghahremani, KL Calkins, HH Wu. Self-Gated Radial Free-Breathing Liver MR Elastography: Assessment of Technical Performance in Children at 3T. *J Magn Reson Imaging*. DOI: 10.1002/jmri.29541

2024 **SG Kafali**, S Shih, X Li, GJ Kim, T Kelly, S Chowdhury, S Loong, J Moretz, S Barnes, Z Li, HH Wu. Automated abdominal adipose tissue segmentation and volume quantification on longitudinal MRI using 3D convolutional neural networks with multi-contrast inputs. *Magn. Reson. Mater. Phys. Biol. Med.* DOI: 10.1007/s10334-023-01146-3 [Code]

2022 KM Strobel, **SG Kafali**, S Shih, AM Artura, R Masamed, D Elashoff, HH Wu, KL Calkins. Pregnancies complicated by gestational diabetes and fetal growth restriction: an analysis of maternal and fetal body composition using magnetic resonance imaging. *J Perinatol*. DOI: 10.1038/s41372-022-01549-5

2022 S Shih, **SG Kafali**, KL Calkins, HH Wu. Uncertainty-Aware Physics-Driven Deep Learning Network for Free-Breathing Liver Fat and R2* Quantification using Self-Gated Stack-of-Radial MRI. *Magn Reson Med*. DOI: 10.1002/mrm.29525

2022 BA Barlas, CD Bahadir, **SG Kafali**, U Yilmaz, EU Saritas. Sheared 2D RF Excitation for Off-resonance Robustness and Fat Suppression in Reduced Field-of-View Imaging. *Magn Reson Med*. DOI: 10.1002/mrm.29416

2022 J Story, S Ghahremani, **SG Kafali**, SF Shih, K Kuwuhara, KL Calkins, HH Wu. Using Free-Breathing MRI to Quantify Pancreatic Fat and Investigate Spatial Heterogeneity in Children. *J Magn Reson Imaging*. DOI: 10.1002/jmri.28337

2022 **SG Kafali**, T Armstrong, S Shih, GJ Kim, JL Holtrop, RS Venick, S Ghahremani, BD Bolster Jr., CM Hillenbrand, KL Calkins, HH Wu. Free-Breathing Radial Magnetic Resonance Elastography of the Liver in Children at 3T: A Pilot Study. *Pediatric Radiology*. DOI: 10.1007/s00247-022-05297-8

2021 X Zhong, T Armstrong, C Gao, MD Nickel, F Han, BM Dale, X Li, **SG Kafali**, P Hu, HH Wu, V Deshpande. Accelerated k-Space Shift Calibration for Free-Breathing Stack-of-Radial MRI Quantification of Liver Fat and R_2^*. *Magn Reson Med*. DOI: 10.1002/mrm.28981

2018 **SG Kafali**, T Cukur, EU Saritas. Phase-Correcting Non-Local Means Filtering for Diffusion Weighted Imaging of the Spinal Cord. *Magn Reson Med*. DOI: 10.1002/mrm.27105

PEER-REVIEWED CONFERENCE PAPERS

2026 BN Zeybek, Ö Turgut, J Pan, R Graf, S Starck, D Rueckert, **SG Kafali**. Opportunistic Cardiac Health Assessment: Estimating Phenotypes from Localizer MRI through Multi-Modal Representations. *MICCAI 2026*. DOI: 10.1007/978-3-032-38260-3_38

2026 W Huang, V, N Stolt-Ansó, N Niessen, M Dannecker, **SG Kafali**, S Kurugol, JA Schnabel, D Rueckert. Gabor Primitives for Accelerated Cardiac Cine MRI Reconstruction. *MICCAI 2026*. DOI: 10.1007/978-3-032-38260-3_38

2026 **SG Kafali***, L Felsner, H Eichhorn, AAJ Leth, A Batvinskas, A Datchev, F Klemm, J Aulich, P Leepagorn, R Klinger, D Rueckert, JA Schnabel. Master Class on Reproducibility: A Student Hackathon on Advanced MRI Reconstruction Methods. *BVM Workshop*. DOI: 10.1007/978-3-658-51100-5_31 (*co-first)

2021 **SG Kafali**, S Shih, X Li, S Chowdhury, S Loong, S Barnes, Z Li, HH Wu. 3D Neural Networks for Visceral and Subcutaneous Adipose Tissue Segmentation using Volumetric Multi-Contrast MRI. *IEEE EMBC*. **(Oral Presentation)**

2021 SF Shih, **SG Kafali**, T Armstrong, X Zhong, KL Calkins, HH Wu. Deep Learning-Based Parameter Mapping With Uncertainty Estimation For Fat Quantification Using Accelerated Free-Breathing Radial MRI. *IEEE ISBI*. **(Oral Presentation)**

2020 **SG Kafali**, S Shih, D Ruan, HH Wu. Adaptive Locally Low Rank and Sparsity Constrained Reconstruction for Accelerated Dynamic MRI. *IEEE ISBI*. **(Oral Presentation, Travel Grant Award)**

2016 **SG Kafali**, T Cukur, EU Saritas. Simultaneous Phase-Correction and Denoising For Diffusion-Weighted MRI. *IEEE SIU*. **(Oral Presentation)**

CONFERENCE ABSTRACTS (SELECTED)

2026 S Starck et al., **SG Kafali**. Abdominal Fat Fraction Atlases for Longitudinal Mapping of Liver Fat During Weight Loss: Results from the LION Study. *ISMRM, Cape Town*. **(Oral)**

2026 BN Zeybek et al., **SG Kafali**. Opportunistic Assessment of Cardiac Function and Structure from Routine Localizer MRI. *ISMRM, Cape Town*. **(Oral)**

2026 Y Zhang, **SG Kafali**, D Rueckert, J Pan. KIMRA: K-space–Image Multimodal Representation Alignment for Comprehensive Cardiac Analysis. *ISMRM, Cape Town*. **(Oral)**

2026 W Zhang, **SG Kafali** et al. 3D nnU-Net-Based Automated Segmentation of Abdominal Adipose Tissue in Children using Free-Breathing Dixon MRI. *ISMRM, Cape Town*.

2025 **SG Kafali** et al. Automated Abdominal Adipose Tissue Segmentation and Quantification in Children using Free-Breathing Dixon MRI and 3D Neural Networks. *ISMRM, Hawaii*. **(Oral)**

2025 S Shih, **SG Kafali** et al. Liver Fat Quantification in Infants using Free-Breathing Stack-of-Radial MRI and Phase-Corrected Non-Rigid Motion Compensation. *ISMRM, Hawaii*. **(Oral)**

2024 **SG Kafali** et al. Improving Self-Gated Radial MR Elastography for Free-Breathing Quantification of Liver Stiffness in Children. *ISMRM, Singapore*.

2023 **SG Kafali** et al. Self-Navigated Rapid Radial Free-Breathing Liver MR Elastography: Assessment of Technical Performance in Children at 3T. *ISMRM, Toronto*.

2022 **SG Kafali** et al. Automated Adipose Tissue Segmentation using 3D Attention-Based Competitive Dense Networks and Volumetric Multi-Contrast MRI. *ISMRM, London*. **(Oral, Magna Cum Laude Award)**

2021 KM Strobel, **SG Kafali** et al. Investigating fetal body composition using magnetic resonance imaging. *WSPR*. **(Oral, Abbot David W. Smith Award)**

2020 **SG Kafali** et al. Fully Convolutional Networks for Adipose Tissue Segmentation Using Free-Breathing Abdominal MRI in Healthy and Overweight Children. *ISMRM*. **(Oral, Magna Cum Laude Award)**

2020 **SG Kafali** et al. Assessment of Free-Breathing Radial Magnetic Resonance Elastography in Healthy Children and Children with Liver Disease at 3T. *ISMRM*. **(Oral, Magna Cum Laude Award)**

2017 **SG Kafali**, T Cukur, EU Saritas. Joint Non-local Means Reconstruction for Correction of Phase-Induced Errors in Diffusion Tensor Imaging. *ISMRM, Hawaii*. **(Magna Cum Laude Award)**

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